SmartRoute AI: An Explainable Risk-Aware Deep Reinforcement Learning Framework for Dynamic Last-Mile Delivery Assignment

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***Abstract*—Assigning delivery orders to suitable delivery partners is a difficult step of last-mile logistics because traffic, deadlines, partner availability, distance and operational risk change continuously. This paper presents SmartRoute AI, an explainable and risk-aware Deep Reinforcement Learning (DRL) framework for dynamic last-mile delivery assignment. A DRL agent observes incoming orders, partner availability, traffic conditions, distance, estimated travel time and delivery priority, and learns to select a suitable partner for each order. Risk factors such as congestion, delayed delivery, excessive travel distance and partner workload are included in the decision process. The Hungarian Algorithm is used as a baseline for comparing assignment performance, and an explainable AI layer states the reason for every assignment, for example lower travel time, reduced risk or better workload balance. The framework is trained and evaluated in a software simulation environment and is planned to be assessed on delivery time, assignment accuracy, total distance, workload distribution and risk reduction. SmartRoute AI aims to provide a scalable, adaptive and explainable software solution for last-mile delivery management.**

***Index Terms*—deep reinforcement learning, last-mile delivery, dynamic assignment, explainable AI, risk-aware optimization.**

I. INTRODUCTION

Last-mile delivery is the final stage of the logistics chain, in which goods travel from a local hub to the customer. It is widely regarded as the most costly and the most difficult stage, because it involves many small orders, strict time expectations and an unpredictable road environment [1]. A central operational decision is the assignment problem: when an order arrives, which available delivery partner should serve it? A poor choice increases travel distance, delays the delivery and overloads some partners while others stay idle.

Classical assignment methods, such as the Hungarian Algorithm [2], find an optimal one-to-one matching for a fixed cost matrix. In practice, however, orders arrive over time and traffic conditions change, so a matching computed on a snapshot can become poor within minutes. Reinforcement learning offers a different approach in which an agent learns an assignment policy from experience in a changing environment [3], [4]. This paper proposes SmartRoute AI, a DRL framework that adds two features often missing from learned dispatchers: explicit consideration of operational risk and a layer that explains each assignment in plain language.

A. Problem Statement

Delivery assignment must account for traffic congestion, delivery deadlines, partner availability, distance and operational risk at the same time. Static or rule-based methods such as nearest-partner assignment do not adapt well when conditions change, and learned policies are often treated as black boxes, which makes it difficult for dispatchers to trust or audit their choices. A method is needed that adapts to real-time conditions, takes risk into account and can justify each decision.

B. Objectives

The objectives of SmartRoute AI are as follows. First, to train a DRL agent that assigns incoming orders to suitable delivery partners in a dynamic environment. Second, to include risk factors, namely congestion, delayed delivery, excessive travel distance and partner workload, in the decision process. Third, to use the Hungarian Algorithm as a baseline so that the learned policy can be compared with a well-known optimisation method. Fourth, to provide an explainable AI layer that gives the reason for each assignment. Finally, to reduce delivery delays, travel distance and operational risk while improving partner utilisation, using a simulation environment that needs no physical hardware.

C. Research Gap

Learned routing and dispatch methods concentrate on minimising cost or travel time and rarely treat risk as an explicit part of the decision [5]–[8]. Studies of explainable reinforcement learning describe how to interpret agent behaviour, but they are seldom applied to delivery assignment [11], [12]. Safe reinforcement learning handles constraints and risk in general settings [13], [14], yet it is not usually combined with workload balance and explanation for last-mile dispatch.

To the best of our survey, few frameworks combine dynamic DRL-based assignment, explicit risk awareness, workload balancing, comparison with an optimisation baseline and per-decision explanation. SmartRoute AI is designed to fill this gap.

II. LITERATURE SURVEY

Representative publications on the vehicle routing problem, optimal assignment, reinforcement learning for dispatch, and explainable and safe reinforcement learning are summarised below, together with their limitations and their relation to the proposed work.

Dantzig and Ramser [15] introduced the truck dispatching problem, which became the vehicle routing problem, and it remains the foundation of delivery optimisation. Boysen et al. [1] surveyed last-mile delivery concepts from an operational research view and described the many variants and the pressure created by customer time expectations. These works define the problem but are based on mathematical optimisation that is usually solved offline for known demand.

Kuhn [2] proposed the Hungarian method, which solves the one-to-one assignment problem optimally for a given cost matrix. It is a natural baseline, and it is used in SmartRoute AI for this purpose. Its limitation is that it optimises one snapshot of costs and does not learn how present assignments affect future orders, workload or risk.

Mnih et al. [3] showed that deep Q-networks can learn control policies directly from experience, and Schulman et al. [4] proposed Proximal Policy Optimization, a stable policy-gradient method. Sutton and Barto [16] give the standard theory of reinforcement learning. These methods supply the learning machinery for the DRL agent, but they do not address delivery assignment on their own.

Nazari et al. [5] and Kool et al. [6] applied neural networks and reinforcement learning to vehicle routing and obtained good routes without hand-designed heuristics. Their focus is route construction rather than the continuous assignment of arriving orders to partners, and they do not model risk or explanation.

In ride-hailing, Xu et al. [7] studied large-scale order dispatch by combining learning and planning, and Lin et al. [8] used multi-agent deep reinforcement learning for fleet management. These works are the closest to the dynamic assignment setting and show that learned dispatch can improve platform-level efficiency. They aim at efficiency measures such as response and utilisation and offer little explanation of individual decisions.

García and Fernández [13] surveyed safe reinforcement learning, and Achiam et al. [14] proposed constrained policy optimisation to keep agents within safety limits during learning. Puiutta and Veith [11] and Heuillet et al. [12] reviewed explainability in reinforcement learning. SmartRoute AI draws on both directions: risk enters the reward and the state, and an explanation layer reports the factors behind each assignment.

The surveyed works fall into optimal assignment [2], learned routing and dispatch [3]–[8] and safe or explainable learning [11]–[14]. Each addresses one part of the problem. The proposed framework combines dynamic learning, risk awareness, an optimisation baseline and explanation in one system.

III. SYSTEM ARCHITECTURE

A. Architecture Diagram

The framework has three layers: an input layer that supplies delivery requests, partner status and traffic data, a decision layer that contains the state builder, risk module, DRL agent and Hungarian baseline, and an output layer that produces the assignment, its explanation and performance metrics (Fig. 1).

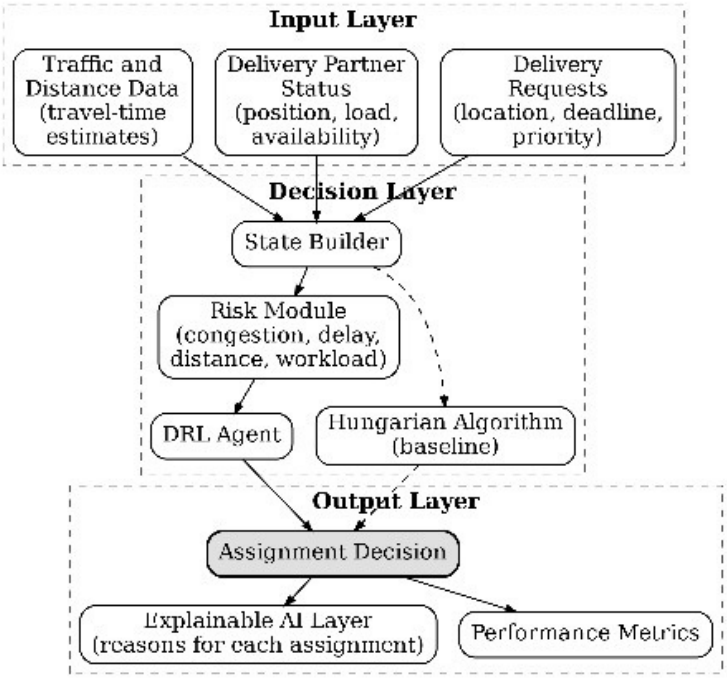


Fig. 1. System architecture of SmartRoute AI.

B. Problem Formulation

The assignment problem is modelled as a Markov decision process. At each decision step t, the state s_t describes the pending orders (location, deadline and priority), each partner (position, current load and availability) and the traffic conditions together with estimated distance and travel time. The action a_t assigns a pending order to an available partner. After the action is executed in the environment, the agent receives a reward r_t and the environment moves to the next state. The agent learns a policy that maximises the expected cumulative reward.

C. Risk-Aware Reward Design

The reward combines the delivery objectives as a weighted penalty, with a bonus for on-time delivery:

$$rt = -(\alpha \cdot D_t + \beta \cdot L_t + \gamma \cdot K_t + \delta \cdot B_t) + \text{bonus for on-time delivery}$$

where D is the delay beyond the deadline, L is the travel distance, K is a combined risk score and B measures workload imbalance across partners. The risk score K is a normalised combination of four factors: traffic congestion on the expected route, probability of a delayed delivery, excess travel distance and the current workload of the selected partner. The weights α, β, γ and δ are tunable, and their effect will be studied experimentally.

D. DRL Agent and Hungarian Baseline

The agent is a neural network that scores each feasible order–partner pair from the state features and selects the best one. A value-based agent such as a deep Q-network [3] and a policy-gradient agent such as PPO [4] are the planned candidates, and the better one will be retained. The Hungarian Algorithm [2] is applied to the same snapshot with a cost matrix built from estimated travel time and risk, and its assignments serve as a reference for assignment accuracy and total cost. A simple nearest-partner rule is added as a second baseline.

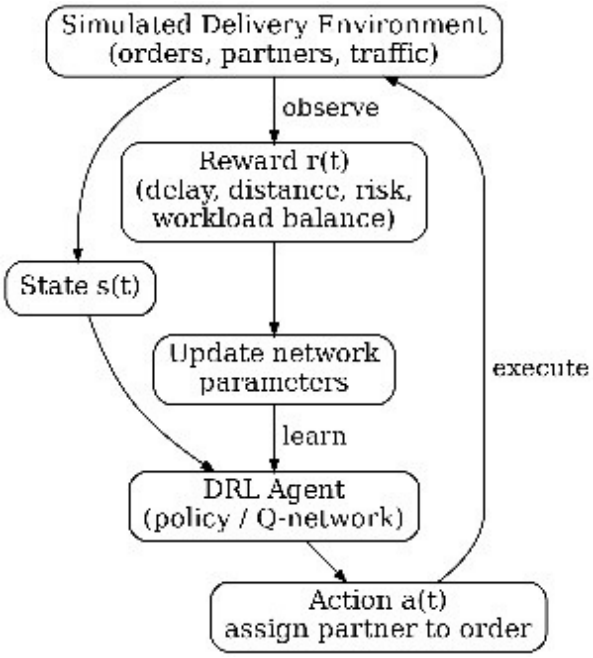


Fig. 2. Training loop of the DRL agent in the simulation environment.

E. Explainable AI Layer

For every assignment, the explanation layer compares the chosen partner with the alternatives on the quantities that drive the decision: estimated travel time, risk score and workload. It then produces a short reason in plain language, such as lower travel time, reduced risk or better workload balance. Because the reward is built from named components, the contribution of each component to the chosen action can be reported directly, and feature-attribution methods such as SHAP [9] and LIME [10] can be applied to the neural network when a finer explanation is needed.

F. Simulation Environment and Planned Implementation

Training and evaluation use a software simulator, so no physical hardware is required. The simulator generates orders over time, places delivery partners on a city map, applies time-varying traffic multipliers to travel times and records deadlines and completions. The environment follows the standard Gym interface [17], and the agent is planned in Python with PyTorch [18] and a reinforcement learning library such as Stable-Baselines3 [19]. All simulated data is labelled as simulated, and the framework is intended to be tested later with real traffic and order data.

IV. EVALUATION PLAN AND DISCUSSION

SmartRoute AI is at the design stage, so this section describes how the framework will be evaluated and discusses its expected strengths and limits. Quantitative results will be reported after the simulation experiments are completed.

A. Evaluation Metrics

Performance will be measured with the criteria in Table I. Each method will be run on the same set of simulated scenarios with several random seeds, and results will be reported as means with their variation, for light, moderate and heavy order loads and for normal and congested traffic.

TABLE I
PLANNED EVALUATION OF THE PROPOSED FRAMEWORK

Criterion	Measure	Purpose
Delivery time	Mean time per order; on-time rate	Speed and reliability
Assignment accuracy	Agreement with Hungarian solution on static snapshots	Quality against optimal baseline
Total distance	~~Distance travelled by all partners~~	Cost efficiency
Workload distribution	~~Spread of orders across partners~~	Utilisation and fairness
Risk reduction	~~Mean risk score of assignments~~	Operational safety
Explanation	~~Reason given for every assignment~~	Transparency

B. Comparison with Existing Approaches

Table II compares SmartRoute AI with the main categories of related work by capability. The comparison is qualitative and not exhaustive.

TABLE II
CAPABILITY COMPARISON BY CATEGORY OF APPROACH

Capability	Hungarian method	Learned routing	Explainable RL	SmartRoute AI
Dynamic adaptation	Partly	Yes	Yes	Yes
Learns from experience	No	Yes	Yes	Yes
Explicit risk factors	No	No	No	Yes
Workload balance	No	No	No	Yes
Per-decision reasons	No	No	Yes	Yes
Optimal baseline	Yes	No	No	Yes

C. Discussion

The expected advantage of SmartRoute AI is that the agent can adapt as traffic and demand change, whereas a snapshot method must recompute from scratch, and that risk and workload are considered together with time and distance. The explanation layer lets a dispatcher see why a partner was chosen, which supports trust and makes it easier to find problems with the learned policy.

Several limitations must be kept in view. The simulator is only an approximation of real traffic and order patterns, so results may change on real data. The reward weights affect the behaviour of the agent and need careful tuning. A learned policy offers no guarantee of optimality, which is why the Hungarian baseline is retained for comparison. These points will be examined during the experiments.

V. CHALLENGES FACED

- Building a simulator that is realistic enough to train a useful policy but simple enough to run many training episodes.

- Choosing the reward weights so that delay, distance, risk and workload balance are traded off sensibly.
- Keeping training stable when orders arrive randomly and traffic conditions change.
- Defining a risk score that is meaningful and can be computed from the available data.
- Producing explanations that are simple for the user yet faithful to the actual decision.

VI. CONCLUSION AND FUTURE WORK

This paper proposed SmartRoute AI, an explainable and risk-aware deep reinforcement learning framework for dynamic last-mile delivery assignment. The framework lets an agent learn to assign orders to delivery partners while considering traffic, distance, deadlines, risk and workload, compares its decisions with the Hungarian Algorithm and explains each assignment in plain language.

Future work includes completing the simulation experiments and reporting the measured results, testing the framework on real order and traffic data, extending it to multiple cooperating agents and combined assignment and routing, and integrating it with a dispatcher dashboard.

REFERENCES

[1] N. Boysen, S. Fedtke and S. Schwerdfeger, "Last-mile delivery concepts: A survey from an operational research perspective," *OR Spectrum*, vol. 43, 2021.

[2] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[3] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, pp. 529–533, 2015.

[4] J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[5] M. Nazari, A. Oroojlooy, L. Snyder and M. Takáč, "Reinforcement learning for solving the vehicle routing problem," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[6] W. Kool, H. van Hoof and M. Welling, "Attention, learn to solve routing problems!," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

[7] Z. Xu *et al.*, "Large-scale order dispatch in on-demand ride-hailing platforms: A learning and planning approach," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2018.

[8] K. Lin, R. Zhao, Z. Xu and J. Zhou, "Efficient large-scale fleet management via multi-agent deep reinforcement learning," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2018.

[9] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[10] M. T. Ribeiro, S. Singh and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016.

[11] E. Puiutta and E. M. S. P. Veith, "Explainable reinforcement learning: A survey," in *Proc. Int. Cross-Domain Conf. Machine Learning and Knowledge Extraction (CD-MAKE)*, 2020.

[12] A. Heuillet, F. Couthouis and N. Díaz-Rodríguez, "Explainability in deep reinforcement learning," *Knowledge-Based Systems*, vol. 214, 2021.

[13] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," *J. Machine Learning Research*, vol. 16, pp. 1437–1480, 2015.

[14] J. Achiam, D. Held, A. Tamar and P. Abbeel, "Constrained policy optimization," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017.

[15] G. B. Dantzig and J. H. Ramser, "The truck dispatching problem," *Management Science*, vol. 6, no. 1, pp. 80–91, 1959.

[16] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[17] G. Brockman *et al.*, "OpenAI Gym," arXiv:1606.01540, 2016.

[18] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[19] A. Raffin *et al.*, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Machine Learning Research*, vol. 22, 2021.